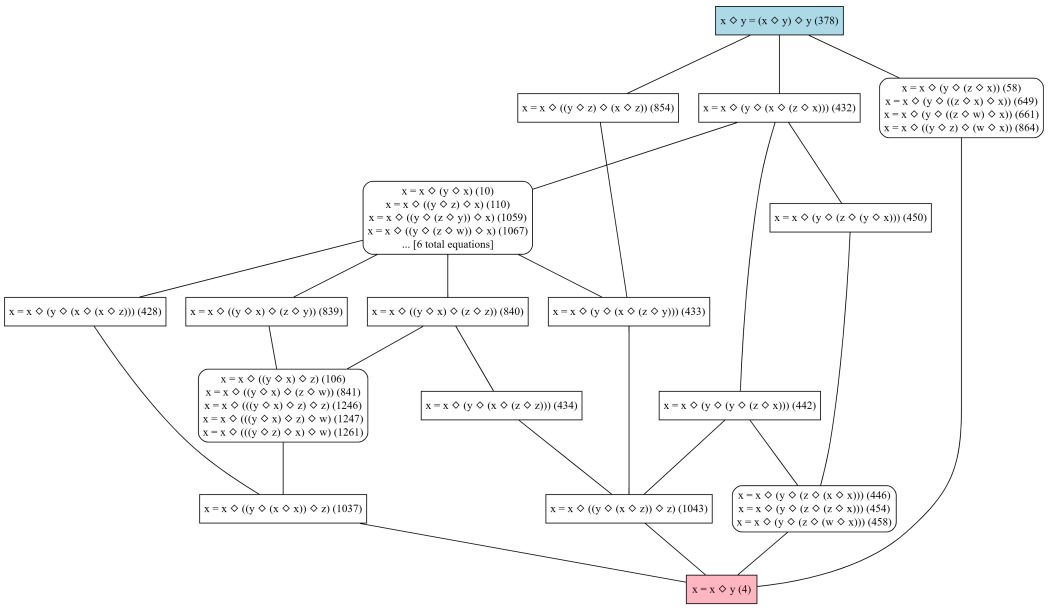

FIGURE 9. Equations similar to E854 that are of the form (11) (possibly involving a fourth indeterminate w) and imply E378. For brevity, 70 equations equivalent to E4 have been omitted.

for some word $f(\mathrm{x}, \mathrm{y}, \mathrm{z})$, that are also known to imply the right-idempotent law E378. An illustrative example is the law E854 depicted in Figure 1. Other examples are listed in Figure 9.

Lemma 6.10. *Equation* E854 *is of the form* (11) *and implies* E378.

Proof. Clearly we have (11) with $f(\mathrm{x}, \mathrm{y}, \mathrm{z}) \coloneqq (\mathrm{y} \diamond \mathrm{z}) \diamond (\mathrm{x} \diamond \mathrm{z})$. From (11) we have in any magma satisfying E854 that

$$x = x \diamond f(x, x, S^2 x) = x \diamond S(x \diamond S^2 x) = x \diamond S(x \diamond f(x, x, x)) = x \diamond Sx.$$

This implies from a further application of (11) that

$$y = y \diamond f(y, x, y) = (y \diamond Sy) \diamond ((x \diamond y) \diamond Sy) = f(x \diamond y, y, Sy)$$

and hence by (11) again

$$(x \diamond y) \diamond y = x \diamond y$$

giving E378. □

Let E be a law of the form (11) that implies E378. We define a directed graph $\to_{\mathrm{E}}$ on words in $\mathcal{M}_X$ by declaring $w' \to_{\mathrm{E}} w$ if $w \sim_{\mathrm{E}} w'' \diamond w'$ for some $w' \in M_X$. By E378 (applied to the quotient magma $\mathcal{M}_{X,\mathrm{E}} = \mathcal{M}_X / \sim_{\mathrm{E}}$), this is equivalent to requiring that $w \sim_{\mathrm{E}} w \diamond w'$. In particular, from (11) we have $f(x, y, z) \to x$ for all x, y, z. Furthermore, the relation $\to_{\mathrm{E}}$ factors through $\sim_{\mathrm{E}}$: if $w \sim_{\mathrm{E}} \tilde{w}$ and $w' \sim_{\mathrm{E}} \tilde{w}'$, then $w' \to_{\mathrm{E}} w$ if and only if $\tilde{w}' \to_{\mathrm{E}} \tilde{w}$.

Call a word $w \in M_X$ *irreducible* if it is not of the form $w = w_1 \diamond w_2$ with $w_2 \to_{\mathrm{E}} w_1$. We can partially understand the equivalence relation $\sim_{\mathrm{E}}$ on irreducible words:

Theorem 6.11 (Description of equivalence)**.** *Let* E *be an equation of the form* (11)*. Let* w *be an irreducible word, and let* w' *be a word with* $w \sim_{\mathrm{E}} w'$*.*

(i) If w *is a product* $w = w_1 \diamond w_2$*, then* w' *takes the form*

$$w' = (\dots((w'_1 \diamond w'_2) \diamond v_1) \diamond \dots) \diamond v_n$$

for some $w'_1 \sim_{\mathrm{E}} w_1$*,* $w'_2 \sim_{\mathrm{E}} w_2$*, some* $n \geq 0$*, and some words* $v_1, \dots, v_n$ *such that for all* $0 \leq i < n$*,* v_{i+1} *is of the form*

$$v_{i+1} \sim_{\mathrm{E}} f(x_i, y_i, z_i)$$

for some x_i, y_i, z_i *with*

$$x_i \sim_{\mathrm{E}} (\dots((w'_1 \diamond w'_2) \diamond v_1) \diamond \dots) \diamond v_i.$$

In particular, $v_{i+1} \to_{\mathrm{E}} x_i$*.*

(ii) Similarly, if $w \in X$ *is a generator of* $\mathcal{M}_X$*, then* w' *takes the form*

$$w' = (\dots(w \diamond v_1) \diamond \dots) \diamond v_n$$

for some $n \geq 0$*, and some words* $v_1, \dots, v_n$ *such that for all* $0 \leq i < n$*,* v_{i+1} *is of the form*

$$v_{i+1} \sim_{\mathrm{E}} f(x_i, y_i, z_i)$$

for some x_i, y_i, z_i *with*

$$x_i \sim_{\mathrm{E}} (\dots(w \diamond v_1) \diamond \dots) \diamond v_i.$$

In particular, $v_{i+1} \to_{\mathrm{E}} x_i$*.*

Conversely, any word of the above forms is equivalent to w*.*

Proof. We just verify claim (i), as claim (ii) is similar. The converse direction is clear from (11) (after quotienting by $\sim_{\mathrm{E}}$), so it suffices to prove the forward claim. By the Birkhoff completeness theorem, it suffices to prove that the class of words described by (i) is preserved by any term rewriting operation, in which a term in the word is replaced by an equivalent term using (11). Clearly the term being rewritten is in w'_1 or w'_2 then the form of the word is preserved, and similarly if the term being rewritten is in one of the v_i. The only remaining case is if we are rewriting a term of the form

$$x_i = (\dots((w'_1 \diamond w'_2) \diamond v_1) \diamond \dots) \diamond v_i.$$

If $i > 0$ we can rewrite this term down to x_{i-1}, and this still preserves the required form (decrementing n by one). If $i = 0$ then we cannot perform such a rewriting because of the irreducibility of $w_1 \diamond w_2$ and hence $w'_1 \diamond w'_2$. Finally, we can rewrite x_i to $x_i \diamond v$ where v is of the form

$$v_i = f(x_i, y, z),$$

and after some relabeling we are again of the required form (now incrementing n by one). This covers all possible term rewriting operations, giving the claim. □

Specializing to the case where w, w' are both irreducible, we conclude

Corollary 6.12 (Unique factorization)**.** *Two irreducible words* w, w' *are equivalent if and only if they are either the same generator of* X*, or are of the form* $w = w_1 \diamond w_2$*,* $w' = w'_1 \diamond w'_2$ *with* $w_1 \sim_{\mathrm{E}} w'_1$ *and* $w_2 \sim_{\mathrm{E}} w'_2$*.*

As an application of this corollary, we establish

Proposition 6.13 (E854 $\not\models$ E3316)**.** *Equation* E854 *does not imply* E3316*.*

Proof sketch. We work in the free magma $\mathcal{M}_X$ on two generators $X = \{\mathrm{x}, \mathrm{y}\}$. It suffices to show that

$$\mathrm{x} \diamond \mathrm{y} \not\sim_{\mathrm{E854}} \mathrm{x} \diamond (\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y})).$$

Suppose this were not the case, then by Corollary 6.12 one of the following statements must hold:

(i) $\mathrm{y} \to_{\mathrm{E854}} \mathrm{x}$.
(ii) $(\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y})) \to_{\mathrm{E854}} \mathrm{x}$.
(iii) $\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y}) \sim_{\mathrm{E854}} \mathrm{y}$.

If (i) holds, then we have $\mathrm{x} \diamond \mathrm{y} = \mathrm{x}$ must hold in $\mathcal{M}_X / \sim_{\mathrm{E}}$, hence E854 would imply E4. However, the magma $\mathbb{Z}/2\mathbb{Z}$ with $x \diamond y = x(1-y)$ refutes this implication.

Similarly, if (iii) held, then E854 would have to imply E10, but this can be refuted by a finite magma with 11 elements.

Finally, if (ii) held, then the claim

$$\mathrm{x} \diamond \mathrm{y} \sim_{\mathrm{E854}} \mathrm{x} \diamond (\mathrm{y} \diamond (\mathrm{x} \diamond \mathrm{y}))$$

to refute simplifies to

$$\mathrm{x} \diamond \mathrm{y} \sim_{\mathrm{E854}} \mathrm{x}$$

and we are back to (i), which we already know not to be the case. □

7. Proof automation

In this project we used proof automation in two ways: automated theorem provers (ATPs) and *Lean* tactics. ATPs are generally stand-alone tools that implement a (semi-)decision procedure for a given formal language or related set of languages. For example, *Vampire* [38, 8] is an ATP focused primarily on first-order logic using superposition, which we used extensively in this project. We also made extensive use of *Prover9* and *Mace4* [43].

ATPs are complex software that can contain bugs. Instead of trusting ATP output, we used proof certificates, which many ATPs can produce, to reconstruct proofs in *Lean*. The details of proof reconstruction depend on the form of the proof certificate produced by the ATP. We expand on this in Section 7.2.

Tactics in *Lean*, on the other hand, are meta-programs [24] that build proofs. In other words, they essentially take *Lean* code as input and produce *Lean* code as output. In this manner, they look like another keyword in the language, and are tightly integrated by producing proofs directly. Under the hood, their implementation can be arbitrarily complex, from syntactic sugar to full decision procedures. The `duper` tactic [17], for example, implements a superposition calculus similar to *Vampire*'s, but for dependent types — *Lean*'s underlying logical foundation.